HWNet: A Compact Temporal CNN for Motor Imagery

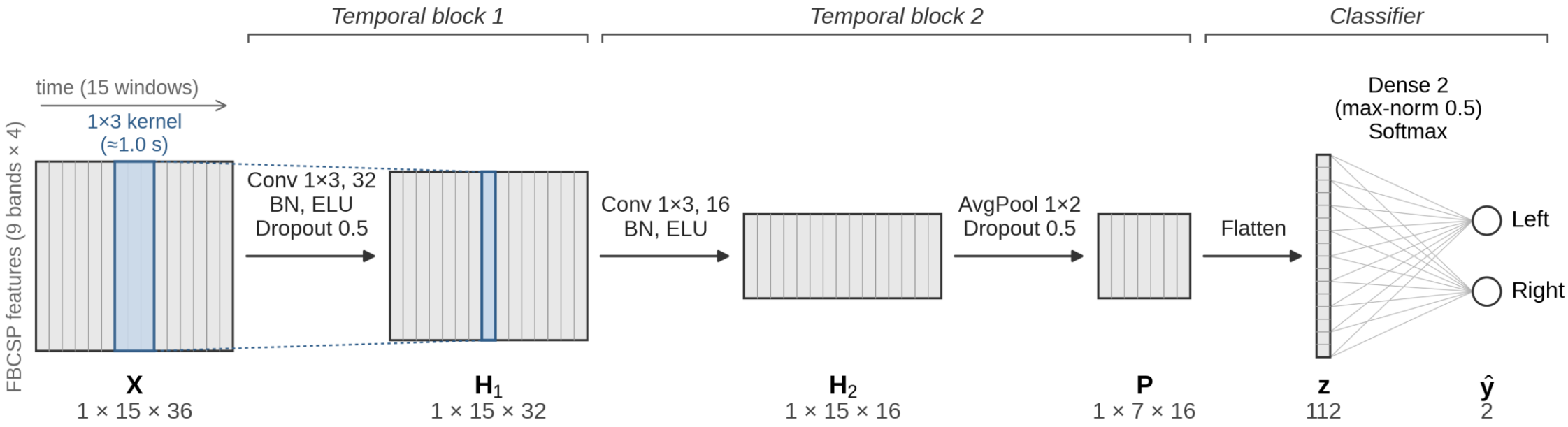
Parts 2–3: Model Architecture and Benchmark

BCI Competition IV-2a, 2-class (left vs. right hand)

Input from Preprocessing (Part 1)

- Each trial arrives as a 36 × 15 feature matrix
  - 9 sub-bands (4–40 Hz) × 4 CSP components = 36 features
  - Log-variance in sliding windows (0.5 s, 0.25 s hop) = 15 time steps
- Spatial and spectral filtering is already done
- What is left for the model: the temporal evolution of ERD/ERS
- Constraint: only ~115 training trials per fold, so the model must be small

HWNet Architecture



- Two temporal conv blocks: 1×3 kernels slide over time only (≈1.0 s per kernel)
- Regularisation: dropout 0.5 (×2), batch norm, max-norm 0.5
- 5,362 trainable parameters, trained with plain cross-entropy

HWNet vs. MixNet

	MixNet (baseline)	HWNet (ours)
Input	Raw EEG, 1 × 400 × 18	Windowed FBCSP power, 1 × 15 × 36
Trainable parameters	44,777	5,362 (~8× fewer)
Loss	MSE + triplet + cross-entropy	Cross-entropy only
Temporal kernels	64 and 32 samples	3 windows (≈1.0 s)
Training time, 9 subjects (GPU)	~18 min	~4.5 min

- Preprocessing already extracts the spatial-spectral information, so the autoencoder and triplet loss in MixNet are not needed
- Fewer parameters and a single loss: nothing to balance or tune

Results on Held-Out Session E

Model	Accuracy (mean ± SD)	Macro F1 (mean ± SD)	Trainable params
MixNet	76.53% ± 15.10%	0.753 ± 0.164	44,777
HWNet	77.13% ± 15.02%	0.757 ± 0.174	5,362

- Protocol: 5-fold CV on Session T, tested on unseen Session E, 9 subjects
- HWNet: +0.6 pp accuracy, better on 4 of 9 subjects
- Difference is not significant (paired t-test p = 0.787, Wilcoxon p = 0.734)
- Takeaway: same accuracy as MixNet with ~8× fewer parameters and 4× faster training

Per-Subject Observations

- Largest gain on S5: 76.81% vs. 60.83% (+16.0 pp)
  - S5 shows a brief, late ERD; the 0.5 s windows likely capture it better than MixNet's long kernels
- S8: 93.47% vs. 90.42%, best subject in the cohort
- S3: 92.22% vs. 91.94%, near ceiling for both models
- S2 is near chance for both models (HWNet 49.44%, MixNet 55.14%)
  - Consistent with BCI inefficiency: no clear μ/β desynchronisation
  - Excluding S2, HWNet mean accuracy is 80.59%

Conclusion

- Windowed FBCSP features (36 × 15) keep the temporal ERD/ERS information in a compact form
- A 2-block temporal CNN with 5.4k parameters is enough on top of these features
- Matches MixNet on Session E (77.13% vs. 76.53%, n.s.) with ~8× fewer parameters
- Faster to train (~4.5 min for all subjects) and no auxiliary losses to tune